

The mixture-of-codings single-effect regression

An empirical-Bayes formulation in the style of fSuSiE:
marginal likelihoods, Bayes factors, and posterior summaries

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Abstract

We give a self-contained, exact empirical-Bayes (EB) derivation of the *mixture-of-codings* single-effect regression (*mixSER*), the single-effect module consumed by the SuSiE / SuSiE 2.0 iterative Bayesian stepwise selection (IBSS) loop. The model is the single-effect regression (SER) of Wang et al. (2020) in which the one causal variant acts through a *coding model* drawn from a finite mixture (additive, dominant, recessive). We deliberately follow the classical Stephens-lab EB style used by **ash**, SuSiE, and fSuSiE: every quantity is a marginal likelihood, a Bayes factor, a posterior inclusion probability, or a posterior moment, all in closed form. There is no variational family and no evidence lower bound (no “free energy”). Attaching the coding mixture to the *variant* rather than to each *individual* keeps the marginal likelihood a finite M -term sum and is exactly what makes the EB treatment exact; the per-individual latent-coding variant, by contrast, has an M^n marginal likelihood and is the only place a variational bound would be required (§7). We derive the single-coding Bayes factor, the coding-mixture Bayes factor, the variant-level posterior (PIP) and coding posterior, the EB updates of the prior variances and mixing weights, and the interface to SuSiE 2.0; the routine reduces exactly to the standard SER when $M = 1$.

1 Model and notation

We mirror the notation of the functional single-effect regression (SFR) model of fSuSiE. Index individuals by $i = 1, \dots, n$, candidate variants by $j = 1, \dots, J$, and coding models by $m = 1, \dots, M$ (typically $M = 3$: additive, dominant, recessive). For variant j under coding m let $x_j^{(m)} = (x_{1j}^{(m)}, \dots, x_{nj}^{(m)})^\top \in \mathbb{R}^n$ be the genotype encoding of that variant under that coding (additive $g \in \{0, 1, 2\}$, dominant $\mathbf{1}\{g \geq 1\}$, recessive $\mathbf{1}\{g = 2\}$). Let $y \in \mathbb{R}^n$ be the response; inside SuSiE this is the partial residual for the single effect being updated. The residual variance σ^2 is fixed externally and updated by the outer IBSS loop.

Covariates, centring, scaling. As in fSuSiE we first project out an intercept and any nuisance covariates Z (sex, age, PCs) from both y and each coding vector, $x_j^{(m)} \leftarrow x_j^{(m)} - Z(Z^\top Z)^{-1} Z^\top x_j^{(m)}$ and likewise for y , and then centre and scale each $x_j^{(m)}$ to unit sample variance. After this reduction the working response and designs are centred, so no intercept term appears below; this is the standard reduction used for univariate regression and is the same one used throughout SuSiE and fSuSiE.

The single-effect regression with a coding mixture. The SER assumes exactly one variant carries a nonzero effect. We encode this with a one-hot selection vector $\gamma = (\gamma_1, \dots, \gamma_J)$ and, for the selected variant, a coding label $c \in \{1, \dots, M\}$:

$$\gamma \sim \text{Mult}(1, \mathbf{p}), \quad \mathbf{p} = (p_1, \dots, p_J), \quad \sum_j p_j = 1, \quad (1)$$

$$c | \gamma_j = 1 \sim \text{Categorical}(\boldsymbol{\pi}), \quad \boldsymbol{\pi} = (\pi_1, \dots, \pi_M), \quad \sum_m \pi_m = 1, \quad (2)$$

$$\beta | c = m \sim \mathcal{N}(0, \sigma_{0m}^2), \quad (3)$$

$$y | \gamma_j = 1, c = m, \beta \sim \mathcal{N}(x_j^{(m)} \beta, \sigma^2 I_n). \quad (4)$$

Thus the prior on the causal variant's effect "channel" is the finite mixture

$$g(\cdot; \text{variant } j) = \sum_{m=1}^M \pi_m \mathcal{N}(0, \sigma_{0m}^2) \text{ acting through the coding-}m \text{ design } x_j^{(m)}, \quad (5)$$

the direct analogue of the fSuSiE scale-mixture prior $g_{sl} = \sum_k \pi_k \mathcal{N}(0, \sigma_k^2)$: the components are indexed by coding rather than by a variance grid, and each component additionally swaps the design vector. A coding with $\sigma_{0m}^2 = 0$ is a point mass at zero and plays the role of the **ash** null component. The unknown hyperparameters are $\mathbf{p}$ (variant prior, usually uniform $1/J$), the mixing weights $\boldsymbol{\pi}$, and the prior variances $\{\sigma_{0m}^2\}$.

Marginal likelihood of the SER. Because γ is one-hot and β is Gaussian, the marginal likelihood is a finite sum of one-dimensional Gaussian integrals,

$$p(y | \mathbf{p}, \boldsymbol{\pi}, \{\sigma_{0m}^2\}, \sigma^2) = \sum_{j=1}^J p_j \sum_{m=1}^M \pi_m \underbrace{\int \mathcal{N}(y; x_j^{(m)} \beta, \sigma^2 I_n) \mathcal{N}(\beta; 0, \sigma_{0m}^2) d\beta}_{=: \text{ML}_{jm}(\sigma_{0m}^2)}. \quad (6)$$

Let $\text{ML}_0 := \mathcal{N}(y; 0, \sigma^2 I_n)$ be the null marginal likelihood (no effect). Every object the routine returns is built from the single-coding marginal likelihoods ML_{jm} and their ratios to ML_0 .

2 Single-coding marginal likelihood and Bayes factor

Fix a variant j and a coding m and write $x := x_j^{(m)}$ for the (centred, scaled) design and $V := \sigma_{0m}^2$. The summary statistics are the usual single-effect ordinary-least-squares estimate and its variance,

$$\hat{\beta} = \frac{x^\top y}{x^\top x}, \quad \hat{s}^2 = \frac{\sigma^2}{x^\top x}, \quad (7)$$

so $\hat{\beta} | \beta \sim \mathcal{N}(\beta, \hat{s}^2)$ is an exact one-dimensional summary of the data for this effect. Completing the square in the Gaussian integral (Appendix A) gives the exact single-effect Bayes factor of Wang et al. (2020) against the null $\beta = 0$,

$$\boxed{\log \text{BF}_{jm}(V) = \log \frac{\text{ML}_{jm}(V)}{\text{ML}_0} = \frac{1}{2} \log \frac{\hat{s}^2}{V + \hat{s}^2} + \frac{\hat{\beta}^2}{2} \left(\frac{1}{\hat{s}^2} - \frac{1}{V + \hat{s}^2} \right)}, \quad (8)$$

with the convention $\log \text{BF}_{jm}(0) = 0$. The posterior of the effect under this single coding is Gaussian, $\beta | (\gamma_j = 1, c = m, y) \sim \mathcal{N}(\mu_{jm}, s_{jm}^2)$, with

$$s_{jm}^2 = \left(\frac{1}{V} + \frac{x^\top x}{\sigma^2} \right)^{-1} = \left(\frac{1}{V} + \frac{1}{\hat{s}^2} \right)^{-1}, \quad \mu_{jm} = \frac{s_{jm}^2}{\sigma^2} x^\top y = s_{jm}^2 \frac{\hat{\beta}}{\hat{s}^2}. \quad (9)$$

Equations (8)–(9) are the additive SER building block: they are computed once per coding and reused everywhere below.

3 Coding-mixture Bayes factor and the variant Bayes factor

Integrating (2)–(3) over the coding label collapses the M single-coding marginal likelihoods into one variant-level Bayes factor,

$$\text{BF}_j = \frac{p(y | \gamma_j = 1)}{\text{ML}_0} = \sum_{m=1}^M \pi_m \text{BF}_{jm}(\sigma_{0m}^2), \quad \log \text{BF}_j = \text{logsumexp}_m(\log \pi_m + \log \text{BF}_{jm}). \quad (10)$$

This is the exact mixture-of-codings Bayes factor: a π -weighted average of the per-coding Bayes factors, evaluated stably by the log-sum-exp identity. It is the single number the IBSS loop needs from the effect, and it specialises to the ordinary SER log-Bayes-factor when $M = 1$.

4 Posterior summaries

Two posterior layers follow from Bayes' rule applied to (1)–(4); both are exact.

Variant posterior (PIP / α). The posterior probability that variant j is the causal one is the SuSiE α vector, equivalently the fSuSiE posterior inclusion probability,

$$\alpha_j := \Pr(\gamma_j = 1 | y) = \frac{p_j \text{BF}_j}{\sum_{j'=1}^J p_{j'} \text{BF}_{j'}} = \text{softmax}_j(\log p_j + \log \text{BF}_j). \quad (11)$$

Coding posterior (responsibilities). Given that variant j is causal, the posterior over which coding it acts through is the ash-style posterior mixture weight

$$w_{jm} := \Pr(c = m | \gamma_j = 1, y) = \frac{\pi_m \text{BF}_{jm}}{\sum_{k=1}^M \pi_k \text{BF}_{jk}} = \text{softmax}_m(\log \pi_m + \log \text{BF}_{jm}). \quad (12)$$

These are the responsibilities, but obtained here in one shot by Bayes' rule at the variant level rather than by iterating a per-individual assignment; they are posterior weights on coding models, not on individuals.

Effect posterior and fitted values. The full posterior of the effect is the two-level mixture: variant j with probability α_j , then coding m with probability w_{jm} , then $\mathcal{N}(\mu_{jm}, s_{jm}^2)$. The posterior-mean contribution of the single effect to the fitted response is

$$\hat{y} = \sum_{j=1}^J \alpha_j \sum_{m=1}^M w_{jm} \mu_{jm} x_j^{(m)}, \quad (13)$$

and, because the design depends on the coding, the SuSiE bookkeeping carries the coding-resolved first and second moments

$$\mathbb{E}[\beta | \gamma_j = 1, y] = \sum_m w_{jm} \mu_{jm}, \quad \mathbb{E}[\beta^2 | \gamma_j = 1, y] = \sum_m w_{jm} (\mu_{jm}^2 + s_{jm}^2), \quad (14)$$

together with the per-coding pair (μ_{jm}, s_{jm}^2) and the design $x_j^{(m)}$ needed to form the expected residual sum of squares in the outer loop. A pointwise $1 - \rho$ credible interval for the effect under variant j is read off the mixture (14) exactly as in `ash`.

5 Empirical-Bayes estimation of the hyperparameters

Here the classical EB treatment replaces the variational M-steps: each hyperparameter is set by maximising a marginal likelihood, never an evidence lower bound.

5.1 Per-coding prior variances σ_{0m}^2

Each σ_{0m}^2 is estimated by maximising its own single-coding marginal likelihood — equivalently, by maximising the per-coding Bayes factor (8) over the prior variance,

$$\boxed{\widehat{\sigma_{0m}^2} = \arg \max_{V \geq 0} \log \text{BF}_{jm}(V), \quad \widehat{\sigma_{0m}^2} = 0 \text{ if } \max_V \log \text{BF}_{jm}(V) \leq 0 \text{ (off-switch).}} \quad (15)$$

This is a smooth one-dimensional problem (solved by Brent’s method) whose objective uses all n individuals, so σ_{0m}^2 is well identified. It is exactly the `estimate_prior_variance` / EBNM step of SuSiE applied coding by coding, and the off-switch is the `ash` null component: a coding whose maximum-marginal-likelihood prior variance is zero is shrunk to the point mass and contributes $\text{BF}_{jm} = 1$ to (10).

5.2 Coding-mixture weights π

The weights π are a shared, locus/architecture-level hyperparameter, and this is the one place where the level of EB pooling matters. As a function of π at a *single locus*, the relevant log marginal likelihood is $\log \sum_m \pi_m \text{BF}_m$, which is concave but maximised at a *vertex* of the simplex — the single coding with the largest BF_m . Point-estimating π from one locus (e.g. running `mixsqp` on a single locus’s coding Bayes factors) therefore collapses all mass onto the sparsest coding; this is the spurious “recessive collapse” pathology. The marginal likelihood identifies π only when many independent loci share it. The correct EB estimate pools the locus-level coding-Bayes-factor vectors $\text{BF}_{\ell} = (\text{BF}_{\ell 1}, \dots, \text{BF}_{\ell M})$ across loci $\ell = 1, \dots, L$,

$$\boxed{\widehat{\pi} = \arg \max_{\pi \in \Delta_M} \sum_{\ell=1}^L \log \sum_{m=1}^M \pi_m \text{BF}_{\ell m},} \quad (16)$$

which is the standard `ash/mixsqp` mixture-weight problem (concave in π , interior optimum) and yields the empirical-Bayes prior shared by all loci. Within any single SER call, π is held fixed at this pooled estimate, or at the uniform default $\pi_m = 1/M$. This places `mixsqp` at the level where it is well posed — across loci, where there are L independent observations — rather than within a locus, where it is degenerate. The picture is identical to fSuSiE: there `ash` pools the T wavelet coefficients of a single effect to estimate π ; here a single locus supplies only M coding Bayes factors, so the pooling must be across loci.

6 mixSER as a SuSiE 2.0 single-effect module

Given $(y, \{x_j^{(m)}\}, \sigma^2, \mathbf{p}, \boldsymbol{\pi})$, one call to mixSER returns, for each variant j : the variant log-Bayes-factor $\log \text{BF}_j$ (10); the posterior inclusion probability α_j (11); the coding posterior $\{w_{jm}\}$ (12); the per-coding posterior moments (μ_{jm}, s_{jm}^2) (9); and the posterior first/second moments (14) with their designs for residualisation. The SuSiE 2.0 IBSS loop uses these exactly as it uses an ordinary SER: for each of the L single effects it forms the partial residual by subtracting the posterior-mean fits (13) of the other effects, calls mixSER, and updates σ^2 from the expected residual sum of squares built from (14). Credible sets and PIPs are assembled from the $\boldsymbol{\alpha}$ vectors as in SuSiE. When $M = 1$ the coding mixture is trivial, $w_{j1} \equiv 1$ and $\text{BF}_j = \text{BF}_{j1}$, so mixSER *is* the standard single-effect regression: it is a strict, drop-in generalisation.

7 Why this stays exact: no free energy

The entire development above is exact because the marginal likelihood (6) is a finite sum: the effect β is integrated in closed form (a one-dimensional Gaussian integral, Appendix A) and the coding label c contributes a single M -way sum per variant. Consequently Bayes factors, posteriors, and the EB updates are all available in closed form, and the method lives entirely within the Stephens empirical-Bayes tradition of `ash`, `SuSiE`, and `fSuSiE`, with no variational family and no evidence lower bound.

Remark 1 (Where a variational bound would have been forced). The exactness hinges on attaching the coding mixture to the *variant*. If instead each individual i carried its own coding label z_i , the marginal likelihood would be $\sum_{z_1, \dots, z_n} \prod_i \pi_{z_i} \int \prod_i \mathcal{N}(y_i; x_{ij}^{(z_i)} \beta_{z_i}, \sigma^2) \prod_m \mathcal{N}(\beta_m; 0, \sigma_{0m}^2) d\boldsymbol{\beta}$, an M^n sum that couples the assignments with the M shared effects and is intractable. That per-individual model is the only one that requires a mean-field variational (free-energy) approximation. The variant-level coding mixture used here removes that obstacle while keeping the feature of interest — allowing the locus to act through additive, dominant, or recessive coding — and is the formulation aligned with `fSuSiE`.

A The single-coding Gaussian integral

For a centred design $x \in \mathbb{R}^n$, centred response y , residual variance σ^2 , and prior $\beta \sim \mathcal{N}(0, V)$, the marginal likelihood is the Gaussian

$$\text{ML}(V) = \int \mathcal{N}(y; x\beta, \sigma^2 I_n) \mathcal{N}(\beta; 0, V) d\beta = \mathcal{N}(y; 0, \sigma^2 I_n + V x x^\top). \quad (17)$$

By the matrix-determinant lemma and the Sherman–Morrison identity, with $\hat{\beta} = x^\top y / x^\top x$ and $\hat{s}^2 = \sigma^2 / x^\top x$,

$$\det(\sigma^2 I_n + V x x^\top) = \sigma^{2n} \left(1 + \frac{V x^\top x}{\sigma^2} \right) = \sigma^{2n} \frac{V + \hat{s}^2}{\hat{s}^2}, \quad (18)$$

$$y^\top (\sigma^2 I_n + V x x^\top)^{-1} y = \frac{y^\top y}{\sigma^2} - \frac{V (x^\top y)^2}{\sigma^2 (\sigma^2 + V x^\top x)} = \frac{y^\top y}{\sigma^2} - \hat{\beta}^2 \left(\frac{1}{\hat{s}^2} - \frac{1}{V + \hat{s}^2} \right). \quad (19)$$

Substituting into $\log(\text{ML}(V)/\text{ML}_0)$ and cancelling the common $y^\top y / \sigma^2$ term gives (8). Completing the square in β in the exponent of $\mathcal{N}(y; x\beta, \sigma^2 I_n) \mathcal{N}(\beta; 0, V)$ gives the Gaussian posterior with precision $1/V + x^\top x / \sigma^2$ and mean $s^2 x^\top y / \sigma^2$, i.e. (9).

References

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